

sophisticated numerical methods. The ability to "think" through problems systematically has opened new possibilities for theoretical economic work. Korinek (2024) provides a guide to how these capabilities can benefit economic research.

The third paradigm, agentic chatbots, emerged in December 2024. It synthesizes fluent language generation, reasoning and autonomous action. Agentic systems can plan sequences of operations, actively gather information, and adapt their strategies based on what they discover. For economists, this means AI systems that don't just respond to questions but actively investigate them—a shift from reactive tools to proactive research partners.

Breakthrough Agents: Deep Research and Claude Code Perhaps no capability better illustrates the potential (and limitations) of AI agents for research than Deep Research systems. As described in Section 3, these systems employ multi-agent architectures to process and synthesize hundreds of sources in minutes, producing comprehensive research reports with accurate citations. When given a research question, an orchestrator agent decomposes it into subtasks, spawns specialized agents to investigate different aspects in parallel, and compiles their findings into a coherent narrative. While these systems compile existing knowledge rather than generating novel insights, they speed up an important part of the research process by automating the time-intensive work of information gathering and initial synthesis. Literature reviews that once took weeks can now be completed in under an hour. However, these systems compile existing knowledge as it is available on the internet and struggle at times to identify the most impactful papers on a given question, especially in less established areas of literature—a critical skill for researchers who work at the frontier.

Anthropic's Claude Code was released as a coding agent in February 2025 to create sophisticated software through natural language descriptions—a practice known as "vibe coding." Many of the examples in this paper were generated with this tool, in minutes rather than hours. However, Claude Code's usefulness extends beyond coding to other areas of cognitive work, including research as it offers an agent that can automate complex workflows simply from natural language descriptions. By some measures, Claude Code has emerged as the most powerful agent in recent months. I felt that using it gave me a glimpse of the future in which more and more cognitive work will be automated.

Democratizing Technical Implementation A striking development documented throughout this paper is how AI agents are making sophisticated technical work accessible to researchers without significant programming expertise. Vibe-coding offers a new paradigm for any research tasks that benefit from computer code. Section 3.3.3 demonstrates how to use vibe-coding to build complete econometric tools from simple English descriptions, handling everything from file uploads to regression analysis to visualization.

This democratization extends beyond simple scripts. Researchers can now build custom data pipelines, implement complex estimation procedures, and create interactive research tools without writing a single line of code. The examples in this paper show agents not just writing code but debugging it, adding features based on feedback, and producing well-documented, professional-quality software. For a discipline where computational methods are increasingly central but programming skills remain unevenly distributed, some suggest that this may lead to a leveling of the technical playing field. However, although all may benefit, a common alternative perspective is that those power users who